

FLOWCORE

Vendor-Agnostic Logical Data Model

AI-Native Data Center Monitoring Platform

Platform Architecture · Entity Model · Source Mapping · ML Pipeline

Version 1.0 | Classification: Confidential



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Why Vendor-Agnostic?

The Problem

- ✗ Previous model hard-coded SNMP OID, Redfish ID, and sensor column names as typed table columns
- ✗ Single dcim_source_id field breaks multi-DCIM environments (nlyte + Sunbird on same floor)
- ✗ Fixed ROOM→ROW→RACK hierarchy cannot model cage, pod, or aisle-level organisations
- ✗ Hard-coded metric columns (cpu_util_pct, power_w) require schema migration for each new source
- ✗ source_protocol as free string breaks entity resolution across vendors using different labels

Design Principles

Canonical entity classes

entity_class discriminator on one INFRASTRUCTURE_ENTITY table — any asset type maps in

EAV for vendor attributes

ENTITY_ATTRIBUTE (namespace + key + value) — zero schema change for new vendor fields

Flexible location tree

Self-referencing LOCATION with depth_level — room, aisle, cage, pod all supported

Catalogue-driven metrics

METRIC_CATALOGUE + METRIC_SOURCE_MAPPING — new metric = 2 rows, not a migration

Configurable resolution

ENTITY_RESOLUTION_RULE stores matching strategies per tenant — no hard-coded OID logic

Four-Layer Logical Model Architecture



Layer 1

Canonical Physical Registry

TENANT · SITE · DATA_CENTER · LOCATION · INFRASTRUCTURE_ENTITY
ENTITY_ATTRIBUTE · ENTITY_RELATIONSHIP · CAPACITY_SPEC

Neo4j (graph nodes + edges)



Layer 2

Source Normalisation

SOURCE_SYSTEM · SOURCE_RECORD · FIELD_MAPPING
IDENTITY_SIGNAL · ENTITY_RESOLUTION_RULE · SYNC_LOG · DEAD_LETTER

PostgreSQL (relational)



Layer 3

Telemetry & Live State

METRIC_CATALOGUE · METRIC_SOURCE_MAPPING · METRIC_DATAPOINT
METRIC_ROLLUP · METRIC_BASELINE · ALERT_CATALOGUE · ALERT_EVENT · GRAPH_MUTATION_LOG

TimescaleDB (hypertables) + S3



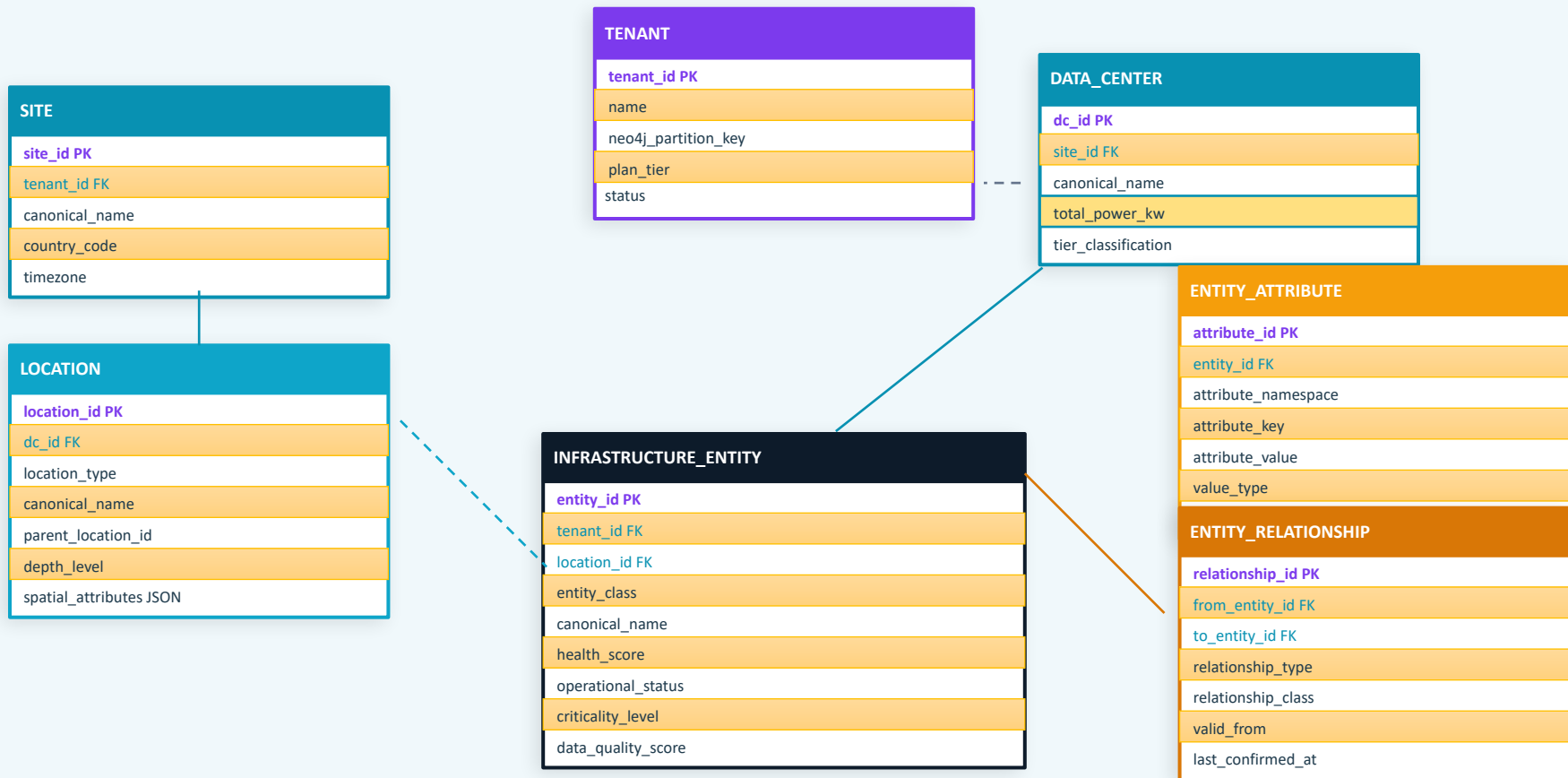
Layer 4

Agent Outputs & Operations

DRIFT_EVENT · CLASSIFICATION_RESULT · CORRELATED_INCIDENT
CAPACITY_FORECAST · ANOMALY_FLAG · SIMULATION_RESULT · AUDIT_LOG

PostgreSQL + Neo4j proposed ns

Canonical Physical Registry — Entity Map



Canonical Physical Registry — Entity Map

CAPACITY_SPEC	
spec_id PK	
entity_id FK	
dimension	
rated_value	
usable_value	
unit	

INFRASTRUCTURE_ENTITY	
entity_id PK	
tenant_id FK	
location_id FK	
entity_class	
canonical_name	
health_score	
operational_status	
criticality_level	
data_quality_score	

ENTITY_RELATIONSHIP	
relationship_id PK	
from_entity_id FK	
to_entity_id FK	
relationship_type	
relationship_class	
valid_from	
last_confirmed_at	

INFRASTRUCTURE_ENTITY — The Universal Asset Record

INFRASTRUCTURE_ENTITY			
entity_id	UUID	PK	Canonical unique ID across all tenants
tenant_id	UUID	FK	Partition key for multi-tenant isolation
location_id	UUID	FK	Link to LOCATION tree node
entity_class	VARCHAR(40)	ENUM	RACK · DEVICE · PDU · FEED · INTERFACE · SENSOR · SERVICE
canonical_name	VARCHAR(200)	NOT NULL	Source-agnostic display name
canonical_type	VARCHAR(80)		Refined type within class (e.g. CRAC within SENSOR)
operational_status	VARCHAR(30)	ENUM	ONLINE · OFFLINE · DEGRADED · MAINTENANCE · UNKNOWN
health_score	FLOAT	0–1	Computed composite score, updated ≤60s lag
active_alert_count	INT		Live count from alert stream, graph property
criticality_level	VARCHAR(20)	ENUM	P1 · P2 · P3 · P4 (maps from any DCIM criticality field)
redundancy_posture	VARCHAR(40)		N · N+1 · 2N · 2N+1 — normalised from DCIM
data_quality_score	FLOAT	0–1	Freshness + completeness composite
last_seen_at	TIMESTAMPZ		Latest telemetry receipt for stale detection
stale_since	TIMESTAMPZ		Set when silence > configurable threshold
created_at	TIMESTAMPZ	NOT NULL	Registry creation timestamp
updated_at	TIMESTAMPZ	NOT NULL	Last property update timestamp

entity_class Values & Mapping

RACK

nlyte: Rack · Sunbird: Cabinet · CSV column: rack_id

DEVICE

DCIM: Device/Asset · SNMP: sysObjectID host · Redfish: Chassis

PDU

DCIM: PowerStrip · IPMI: PDU entity · SNMP: powerMIB

FEED

DCIM: PowerFeed · EcoStruxure: Circuit · CSV: feed_id

INTERFACE

SNMP: ifTable · Redfish: EthernetInterface · DCIM: Port

SENSOR

DCIM: Sensor · MQTT payload type=sensor · IPMI: SDR

SERVICE

DCIM: Service · CMDB: CI · Manual entry

Source Normalisation Layer — Entities

SOURCE_SYSTEM

source_id PK

tenant_id FK

source_name

source_class ENUM

vendor_name

product_version

connection_type ENUM

connection_config JSON

auth_type

status

last_sync_at

source_class: DCIM · TELEMETRY · DISCOVERY · LOG ·
MANUAL

FIELD_MAPPING

mapping_id PK

source_id FK

source_entity_type

source_field_path

canonical_target

canonical_namespace

canonical_key

transform_type

transform_expression

value_type

is_identity_field

is_required

transform_type: DIRECT · UNIT_CONVERT · ENUM_MAP ·
EXPRESSION · CONSTANT

IDENTITY_SIGNAL

signal_id PK

entity_id FK

source_id FK

tenant_id FK

signal_type ENUM

signal_value

match_confidence

resolution_status

observed_at

resolved_at

signal_type: MAC_ADDRESS · SERIAL · IP · HOSTNAME
· OID · RACK_USLOT · ASSET_TAG

ENTITY_RESOLUTION_RULE

rule_id PK · tenant_id FK · rule_name · match_strategy ENUM · match_criteria JSON ·
confidence_threshold · conflict_resolution · priority · is_active

SYNC_LOG

log_id PK · source_id FK · sync_type · sync_mode ENUM · records_received · records_mapped ·
records_rejected · entities_created · entities_updated · rejection_summary · duration_ms · started_at
· completed_at · status

Source Mapping Examples — Any Vendor → Canonical

nlyte DCIM	rack_name	→	INFRASTRUCTURE_ENTITY.canonical_name	DIRECT
	RackMaxPower	→	CAPACITY_SPEC.rated_value (dim=power_kw)	UNIT_CONVERT ÷1000
	CabinetOwner	→	ENTITY_ATTRIBUTE (ns=dcim.nlyte, key=owner)	DIRECT
	CriticalityLevel	→	INFRASTRUCTURE_ENTITY.criticality_level	ENUM_MAP: High→P1
	RackUHeight	→	CAPACITY_SPEC.rated_value (dim=space_u)	DIRECT
Prometheus	node_cpu_seconds_total	→	METRIC_DATAPOINT (metric=cpu_util_pct)	EXPRESSION: rate(5m)×100
	node_memory_MemUsed	→	METRIC_DATAPOINT (metric=mem_util_pct)	UNIT_CONVERT / MemTotal
	ipmi_power_watts	→	METRIC_DATAPOINT (metric=power_draw_w)	DIRECT
	instance label	→	IDENTITY_SIGNAL (type=HOSTNAME)	EXTRACT: label value
SNMP Walk	sysObjectID (.1.3.6.1.2.1.1.2.0)	→	IDENTITY_SIGNAL (type=OID)	DIRECT
	sysDescr (.1.3.6.1.2.1.1.1.0)	→	ENTITY_ATTRIBUTE (ns=snmp, key=sys_descr)	DIRECT → NLP pipeline
	ifOperStatus per row	→	INFRASTRUCTURE_ENTITY (class=INTERFACE)	PER_ROW: one entity per ifIndex
	entPhysicalSerialNum	→	IDENTITY_SIGNAL (type=SERIAL)	DIRECT

Telemetry & Live State Layer — Generic Metric Model

Source System

Prometheus · Zabbix
SNMP · Vendor API

METRIC_SOURCE_MAPPING

source_metric_name
unit_conversion_expr

METRIC_CATALOGUE

canonical_metric_name
unit · alert_thresholds

METRIC_DATAPOINT

entity_id · metric_id
value · event_ts

METRIC_ROLLUP

tenant_id FK
entity_id FK
entity_class
metric_id FK
window_start
window_size ENUM
avg_value
min/max/p50/p95/p99
sample_count
quality_summary

tenant_id FK
window_size: 5m · 1h · 24h — any entity type, any metric

METRIC_BASELINE

baseline_id PK
tenant_id FK
entity_id FK
metric_id FK
mean_value
stddev_value
p5/p95_value
baseline_window_days
sample_count
computed_at
valid_from/to

Recomputed daily by P11. Excludes windows around
GRAPH_MUTATION_LOG events

ALERT_CATALOGUE

alert_type_id PK
tenant_id FK
canonical_alert_name
alert_category
default_severity
entity_class_scope
description
is_active

ALERT_SOURCE_MAPPING

mapping_id PK
alert_type_id FK
source_id FK
source_alert_name
source_severity_map
source_entity_id_field

source_severity_map: JSON e.g. {DISASTER:P1,
HIGH:P2}

Telemetry & Live State Layer — Generic Metric Model

GRAPH_MUTATION_LOG

tenant_id FK

mutation_id PK

operation_type ENUM

entity_class

entity_id

relationship_type

from/to_entity_id

before_state JSON

after_state JSON

source_service

kafka_topic

occurred_at

TimescaleDB materialisation of graph.mutations Kafka topic. Enables time-travel queries for post-incident analysis

ALERT_SOURCE_MAPPING

mapping_id PK

alert_type_id FK

source_id FK

source_alert_name

source_severity_map

source_entity_id_field

source_severity_map: JSON e.g. {DISASTER:P1, HIGH:P2}

Agent Output Entities — All Three Phases



Phase 1 · Topology & Classification

DRIFT_EVENT

drift_type · severity · source_a_id/b_id · conflict_field ·
detection_confidence

DRIFT_SUGGESTION

proposed_action · proposed_namespace_ref · operator_decision ·
decided_by FK

CLASSIFICATION_RESULT

inferred_entity_type · capability_flags JSON · confidence_score ·
needs_review · mlflow_run_id



Phase 2 · Correlation/RCA + Prediction

CORRELATED_INCIDENT

root_cause_entity_id · confidence_score ·
evidence_entity_ids · narrative · operator_verdict ·
is_false_positive

CAPACITY_FORECAST

entity_id FK · metric_id FK · scope_level ·
horizon_label · predicted_value/lower/upper ·
exhaustion_flag

ANOMALY_FLAG

entity_id FK · metric_id FK · observed_value ·
baseline_mean · deviation_sigma · severity

FAILURE_PROBABILITY

entity_id FK · probability_score ·
contributing_factors JSON · horizon_label ·
mlflow_run_id

Agent Output Entities — All Three Phases



Phase 3 · What-If Planning

SIMULATION_SCENARIO

target_entity_id · failure_mode · change_type · chained_steps JSON · is_saved

SIMULATION_RESULT

affected_entity_count · impact_severity_map JSON · redundancy_assessment · graph_snapshot_ref

11 Required Pipelines — All Three Phases

DCIM Ingestion

Scheduled + on-demand
[dcim-ingestion-service](#)

dcim.config.raw/normalized Kafka

P1

Telemetry Gateway

Continuous event-driven
[telemetry-gateway-service](#)

metrics/alerts/events.raw Kafka

P1

Active Discovery

Scheduled per subnet
[active-discovery-service](#)

discovery.snmp/bmc.results Kafka

P1

Digital Twin Construction

Continuous Kafka consumer
[graph-updater-service](#)

Neo4j upsert + graph.mutations

P1

Time-Series Storage

Continuous Kafka consumer
[timeseries-writer + archiver](#)

TimescaleDB METRIC_DATAPOINT + S3

P1

Topology & Classification

Event-driven + full scan
[reconciliation + class agents](#)

drift.events · classification.results

P1

Correlation & Prediction

Continuous + scheduled
[rca + prediction agents](#)

incidents.correlated · forecasts

P2

What-If Simulation

On-demand operator-initiated
[what-if-planning-agent](#)

SIMULATION_SCENARIO/RESULT

P3

ITSM Integration

Continuous Kafka consumer
[eventing-integration-service](#)

ServiceNow API · Jira API

P1

Model Operations

Scheduled + drift breach
[MLflow + retraining DAGs](#)

MLflow registry + model artefacts

P2

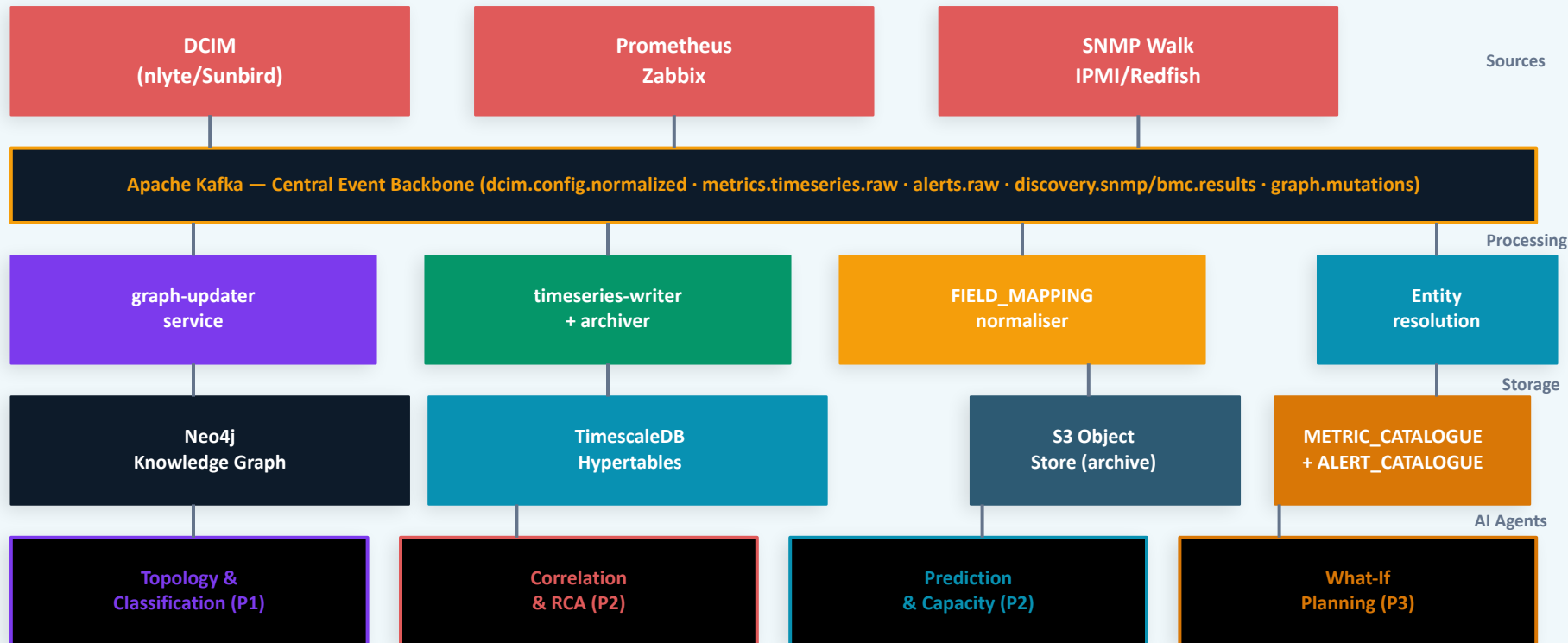
Continuous Aggregation

Automatic (DB-native)
[TimescaleDB cont. aggregates](#)

METRIC_ROLLUP · METRIC_BASELINE

P1

Pipeline Data Flow — Source to AI Agents



10 ML Models — Training Data & Feature Sources

M1 Device type classifier

Algorithm: LightGBM

Train: SNMP results · Redfish · CLASSIFICATION_RESULT feedback

Features: OID enterprise bucket · sys_descr TF-IDF · MIB flags · port profile · Redfish type

P1

M2 Capability inferrer

Algorithm: Rule + NLP

Train: SNMP/Redfish results · operator corrections

Features: OID capabilities · banner string NLP · same feature pipeline as M1

P1

M3 Alert clusterer

Algorithm: DBSCAN

Train: ALERT_TIMESERIES · CORRELATED_INCIDENT feedback

Features: temporal_distance · graph_hop_distance (Neo4j) · alert_semantic_similarity TF-IDF

P2

M4 Root cause scorer

Algorithm: Bayesian network

Train: CORRELATED_INCIDENT confirmed incidents · FAILURE_PROBABILITY priors

Features: upstream_alert_count · first_alert_lag · path_depth · recent_graph_mutation

P2

M5 Capacity forecaster

Algorithm: Prophet + LSTM

Train: RACK/DEVICE_METRIC_ROLLUP (90d min) · CAPACITY_SPEC ceilings

Features: hour_of_day/week · rolling_mean_1h/7d · capacity_headroom · dcim_change_flag

P2

M6 Anomaly detector

Algorithm: STL + Isolation Forest

Train: METRIC_DATAPOINT · METRIC_BASELINE

Features: residual component (STL) · power_per_cpu_ratio · temp_delta_from_baseline

P2

M7 Failure probability

Algorithm: Cox PH survival

Train: DEVICE_METRIC_ROLLUP · CORRELATED_INCIDENT failure events

Features: device_age · avg_inlet_temp_30d · error_rate_trend · anomaly_flag_count · mtbf_ratio

P2

M8 Degradation trend

Algorithm: Linear regression

Train: DEVICE_METRIC_ROLLUP 14d windows

Features: trend_slope · intercept · value_as_pct_of_threshold

P2

M9 Graph impact scorer

Algorithm: LightGBM + Monte Carlo

Train: CORRELATED_INCIDENT.impact_severity_map · resolved incidents

Features: edge_type · path_depth · criticality · service_tier · failure_probability M7

P3

M10 Load redistributor

Algorithm: CSP solver (no training)

Train: No training — parameterised by live METRIC_ROLLUP + CAPACITY_SPEC

Features: current_load · max_capacity · redundancy_paths from Neo4j graph

P3

Shared Feature Store — Preventing Training/Serving Skew

All features are computed identically at training time (from TimescaleDB history) and at inference time (from live METRIC_DATAPOINT). The feature store ensures the same transformation code, versioned alongside each model in MLflow (MG-02).

Device identity → M1, M2, M9	Temporal / metric → M5, M6, M7, M8	Alert / topology → M3, M4, M9
snmp_enterprise_bucket discovery.snmp.results sys_oid prefix → manufacturer enum	hour_of_day / day_of_week Derived from METRIC_DATAPOINT.event_ts	alert_temporal_distance_s abs(event_ts_a - event_ts_b) from ALERT_TIMESERIES
sys_descr_tfidf[50] sys_descr → NLP pipeline → SVD-50 dimensional vector	rolling_mean_1h/6h/24h/7d METRIC_ROLLUP (window_size) per entity per metric	alert_semantic_similarity TF-IDF cosine similarity of canonical_alert_name pairs
mib_support_flags[8] Supported MIB OID set → boolean array (power/UPS/entity/BGP...)	power_per_cpu_ratio Derived: power_draw / (cpu_util + 1) — PSU degradation signal	graph_hop_distance Neo4j shortest path (Cypher) between entity_ids
port_profile[4] total_port_count · ethernet_ratio · fibre_ratio · serial_present	temp_delta_from_baseline inlet_temp_c - METRIC_BASELINE.mean_value	upstream_alert_count Count of ALERT_EVENT downstream of candidate root cause
redfish_type_category discovery.bmc.results @odata.type → enum	error_rate_trend_slope Linear regression slope over 30-day error_count window	first_alert_lag_ms Time from candidate entity alert to cluster first alert
node_criticality INFRASTRUCTURE_ENTITY.criticality_level	anomaly_flag_count_30d COUNT(ANOMALY_FLAG) per entity last 30 days	recent_graph_mutation Boolean: GRAPH_MUTATION_LOG entry on entity in last 24h
	mtbf_age_ratio device_age_days / manufacturer_mtb_days (static reference)	historical_root_cause_count Count of CORRELATED_INCIDENT where entity was root cause
	capacity_headroom_pct Derived from CAPACITY_SPEC rated_value vs current METRIC_ROLLUP	failure_probability M7 FAILURE_PROBABILITY.probability_score — cross-model feature

Governance, RBAC & Audit Architecture

Role-Based Access Control

NOC Operator

Read graph · Submit alert feedback · View drift events

Change Manager

Read graph · Run what-if simulations · Save scenarios

DC Admin

Read + write entity attributes · Approve drift suggestions

Platform Admin

Full access · Manage tenants · Configure source systems

Audit & Compliance

AUDIT_LOG_ENTRY

Every user action, API call, and AI agent decision logged with user_id, timestamp, resource, change payload. is_immutable=true enforces 12-month retention (NFR-14, SEC-01)

Multi-tenant isolation

TENANT.neo4j_partition_key enforces graph partition boundaries. tenant_id on every table. Automated access boundary tests in CI/CD (NFR-13, SEC-03)

Model governance (MLflow)

All ML models versioned in MLflow with training metadata, evaluation metrics, feature set, deployment history. Automated retraining on drift breach (MG-01 to MG-05)

Data quality scoring

NODE_QUALITY_SCORE computed per INFRASTRUCTURE_ENTITY: completeness (fraction of expected ENTITY_ATTRIBUTE rows present) + freshness (last_updated_at vs threshold) = overall_score surfaced in Drift & Hygiene Dashboard (DQ-04)

Dead letter quarantine

DEAD_LETTER_RECORD captures all schema-invalid records with kafka_topic, partition, offset, rejection_reason for replay or manual review. Alerts fire to platform ops team (DQ-01)

Vendor-Agnostic Logical Data Model

4

Model layers

Registry · Normalisation · Telemetry · Agents

30+

Logical entities

Canonical + source-agnostic + agent outputs

11

Pipelines

P1–P6 Phase 1 · P7–P8 Phase 2–3 · P9–P11
ops

10

ML models

Classification · RCA · Forecasting · Simulation

Source-agnostic core

INFRASTRUCTURE_ENTITY + ENTITY_ATTRIBUTE accept any DCIM vendor
— nlyte, Sunbird, Device42, NetBox, CSV

Configurable resolution

ENTITY_RESOLUTION_RULE stores tenant-specific identity matching
strategies — MAC, serial, IP, rack+U-slot, hostname

Catalogue-driven telemetry

METRIC_CATALOGUE + ALERT_CATALOGUE normalise Prometheus, Zabbix,
SNMP, InfluxDB, vendor APIs — no schema changes

Graph + relational + TSDB

Neo4j for topology traversal, TimescaleDB for metric history, PostgreSQL
for operational records — right tool per access pattern